Hildebrand: the acknowledged k=2 cell boundary

Result. Lean refutes the original printed Lemma4 at fixed k=2, ε=1, $m_2=4$, $a_1=0$ and j=1. For every prime p≥3, its actual uniform cell probability is1/(p−1), strictly greater than the printed bound11/(40p). Lean directly rules out any sufficiently-large-prime threshold for this specialization. This is a formalization of an author-acknowledged boundary issue, with no novelty claim or refutation of valid mixing Theorems1/2. [pcc:unverified]

Literal original cell	k=2, ε=1, $m_2=4$, $a_1=0$, j=1; prime p≥3 Canonical centered residue / p lies in [1/p, 5/(4p)]
Actual probability	Full uniform nonzero choices; cell is $a_2=-1$ Probability 1/(p−1) > printed bound 11/(40p)
Scope and repair	No sufficiently-large-prime threshold fixes this cell Author acknowledged k=2 issue; valid mixing retained

Faithful source distribution. Actual nonzero elements of Fp have proved cardinality p−1. A genuine normalized uniform Mathlib PMF is constructed, together with an actual equivalence to all injective two-point supports whose first value is0. Its pushforward is proved uniform on that full conditional space. The actual source expression $(a_1-a_2)\cdot j$ equals$-a_2$. No selected support subclass or supplied probability/cardinality premise is used. [pcc:unverified]

Canonical representatives and exact event. Lean proves congruence, half-open range(−p/2,p/2] and uniqueness for the actual centered representative. The literal real-power cell formulas and decimal1.1 reduce to endpoints1/p and5/(4p), with bound11/(40p). The cell is equivalent to centered residue1, hence exactly the actual choice $a_2=-1$. Uniform expectation gives1/(p−1). Euclid's infinitude of primes closes the direct eventual-bound negation. [pcc:unverified]

Credited repair and positive control. The later author preprint explicitly treats k=2 separately, restricts its counting lemma to k≥3 and flags the older failure as an exercise. The qualified two-support mixing result for any fixed positive normalized weights is included as a Lean positive control, using the preceding actual independent-law coupling proof; it does not depend on this invalid cell bound. Corrected k≥3 counting and mixing Theorems1/2 are retained. [pcc:unverified]

Primary review. Hildebrand1994, PTRF100,191–203: full13 published pages read/visually inspected; original Lemma4p196/proof197 and k≥2 contextp195 rechecked. Author preprint math/0401189v1: selected section4.2/preprint13–16/reference14p36 read, pages15–16 visually inspected; not a full37-page read. Author corrections index found, attachment unread after recorded access failures; acknowledgement comes from the accessible preprint itself. [pcc:unverified]

Checks and remainder. Wholegate30/0failures/0trusted; all54 build/processing/provenance files retained in25objects, fivegate files and recursive source/config pins. Lean4.35.0/Mathlib56a00585…, standard foundations/default limits. Signed standing unverified. Stronger general random-support exponent and other14gaps remain open. KB/Conductor unavailable, fallback recorded before code. No holes/custom axioms/native evaluation/raised limits/global flags/waivers/native experiment/paid compute/upload. [pcc:unverified]